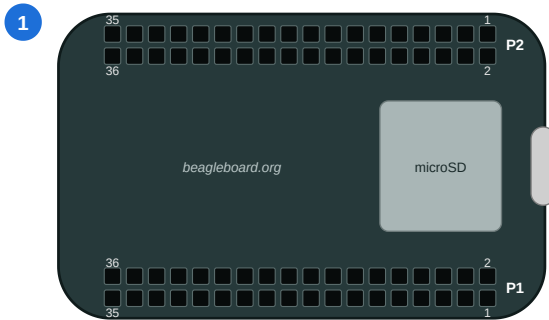


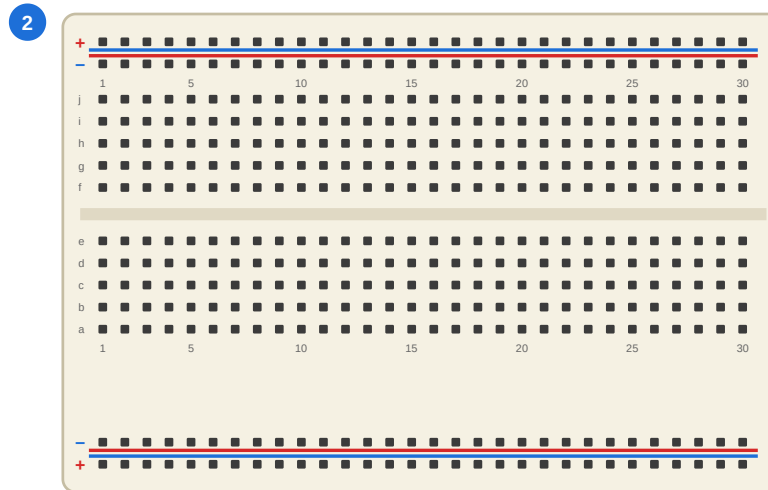
What is in the bag

Tip everything out and match each thing to a picture. Everything here is drawn at the same scale as every later picture.



PocketBeagle 2

The computer. Two black strips of sockets (P1, P2) along the long edges, tiny numbers at the strip ends. Hold it by the edges. The side with the microSD slot and the dog logo is the side you use.



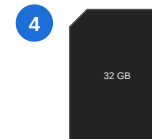
Breadboard, half size

A slab of holes. Red and blue lines mark the power rails on each long edge. A mini board (17 columns, one rail pair) works too.



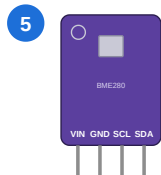
USB-C cable

Must be a data cable. A charge-only cable looks identical and is the most common reason nothing works.



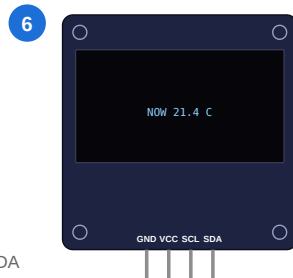
microSD card

8 GB or more. Holds the whole operating system.



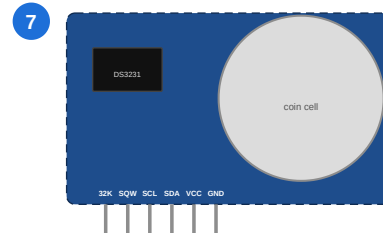
BME280 sensor

Fingernail sized, purple or blue, four pins (a few have six): VIN or VCC, GND, SCL, SDA in tiny letters. The small metal can is the sensor itself. If the pins are loose in the bag instead of soldered on, stop: that is a different job.



OLED screen (SSD1306, 0.96 inch)

A square board with a dark glass window and four pins on one edge: VCC, GND, SCL, SDA. The order of VCC and GND **differs between makers. Read yours now and write it down.** Getting those two backwards is the one mistake in this guide that kills a part instantly.



DS3231 clock (optional)

The one with the round coin cell holder. Six pins; only SCL, SDA, VCC, GND are used. Leave it in the bag on your first build. No battery until you have read the battery note in `HARDWARE.md`.

Jumper wires

Male to male: a pin at both ends. 12 needed, 16 with the clock.



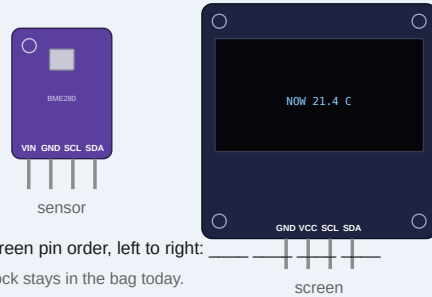
Any four colours work if you use them the same way throughout.

Not in the bag and not needed: a monitor, keyboard or power supply for the board. Your laptop is all three, through the USB-C cable.

Lay it out like this before you start

This is the arrangement every later picture uses. You sit at the bottom edge. Cable and card stay where they are until Parts 3 and 4.

1 Modules, pins toward you

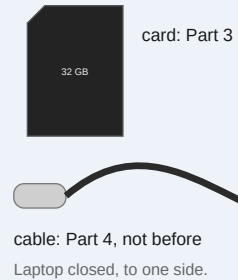


2 Jumpers in four rows, by colour



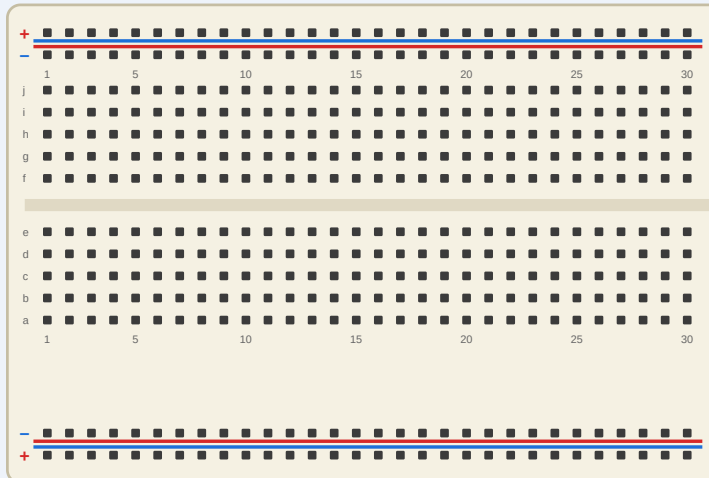
Count them now. Hunting for a missing wire at step 3 costs ten minutes of doubt.

3 Not yet



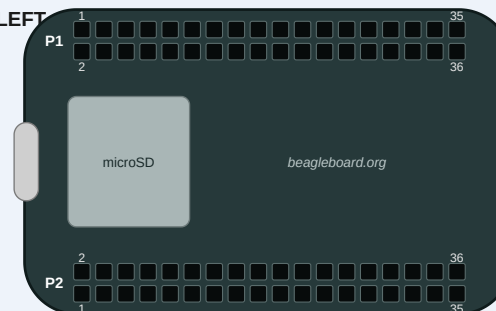
4 Breadboard, the long way across

Modules go along the top edge.
Power uses the rail pair along the bottom edge, nearest the board. If your board has one rail pair only, turn it so that pair is at the bottom.



Rows j to f (top half) hold the modules and the two shared columns. Rows e to a are unused; wires fly over them.

5 PocketBeagle 2, sockets up, USB-C port to the LEFT



Turned so P1 is the strip nearest the breadboard. Pin 1 is then at the left end of P1, by the USB-C port.

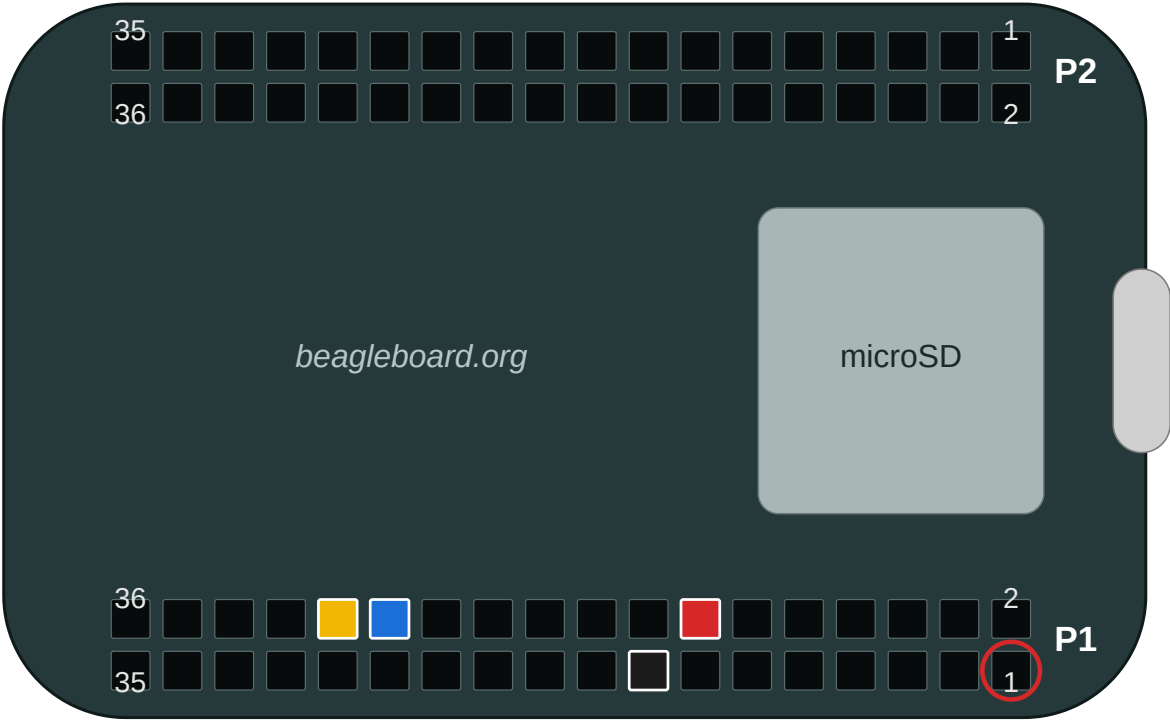
Find the tiny printed 1 and 2 there before going on.

Every wire that leaves P1 goes straight up into the breadboard.

your table, seen from above

Where the four pins are

Board held with the sockets facing you and the USB-C port on the right, as BeagleBoard photographs it. On your table it is turned round; the numbers are the same.



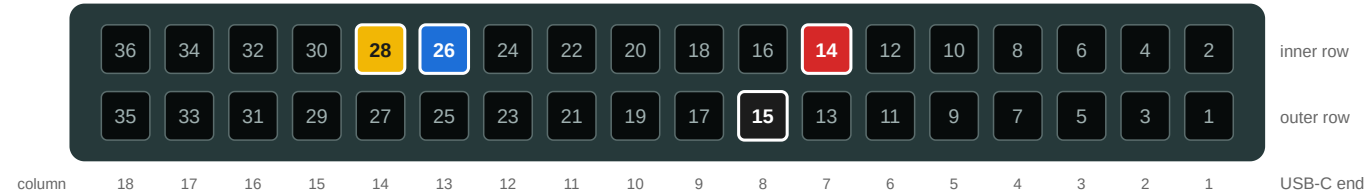
The four we use

- 14** **P1.14 VDD_3V3**
3.3 V out. column 7, inner row, from the USB-C end.
- 15** **P1.15 GND**
ground. column 8, outer row, from the USB-C end.
- 26** **P1.26 I2C2_SDA**
SDA, data. column 13, inner row, from the USB-C end.
- 28** **P1.28 I2C2_SCL**
SCL, clock. column 14, inner row, from the USB-C end.

Trust the numbers printed on the board over this picture. Find the printed 1 and 2 at the USB-C end of P1 and the 35 and 36 at the far end. If they are there, the columns are right.

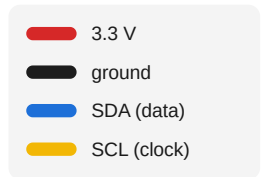
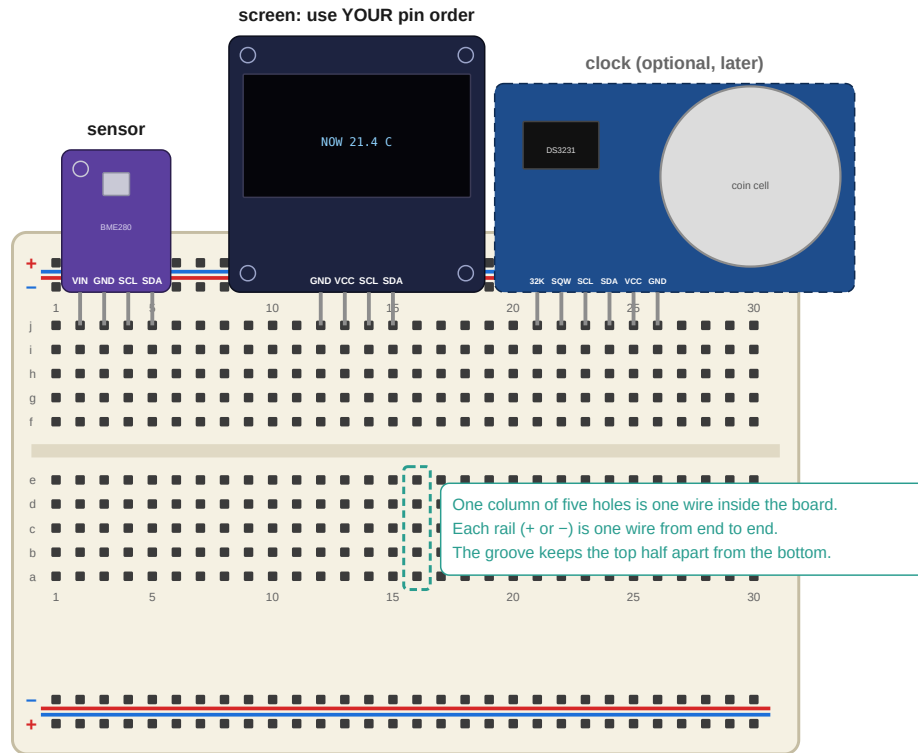
P1.1 is 5 V. Nothing ever goes in here.

P1 close up. Column 1 is at the USB-C end. Outer row = odd pins, inner row = even pins.



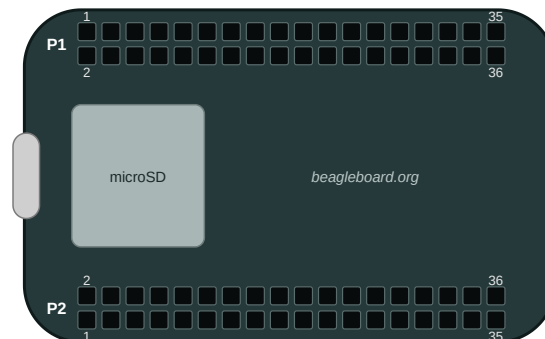
Step 1 of 4: seat the modules

Push each module into the top half so every pin has its own column. The module body sits over the rails. USB cable still in the bag.



board turned so P1 faces the breadboard: USB-C on the left, pin 1 at the left end of P1.

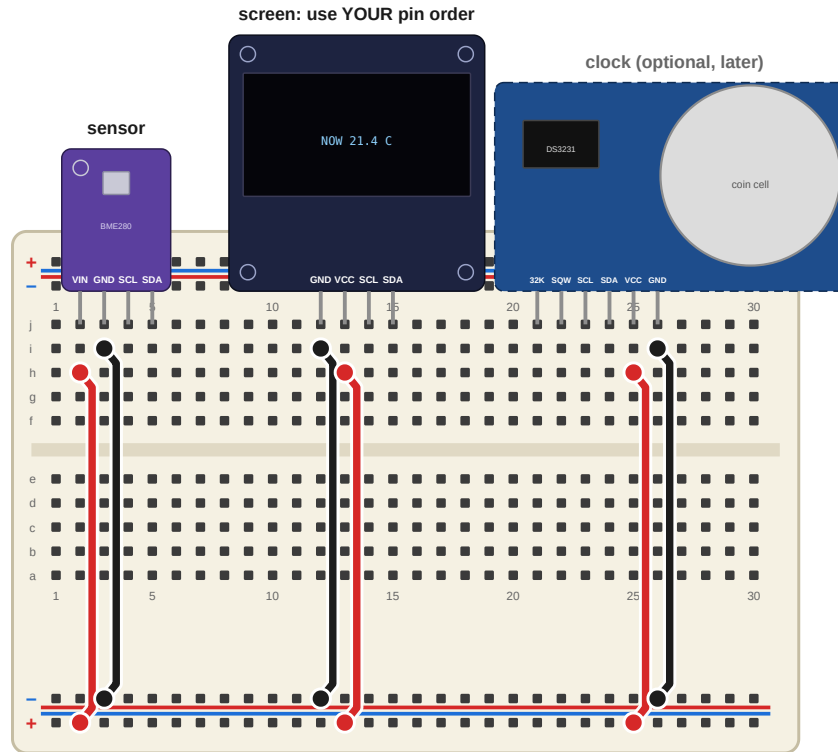
the rails: follow the printed red and blue lines on your own board; which one is outer varies.



Press each module in firmly and evenly; it takes more force than you expect. Pins in row j, one pin per column. The bodies hang over the top edge and cover the top rails, which is why power uses the bottom rails.

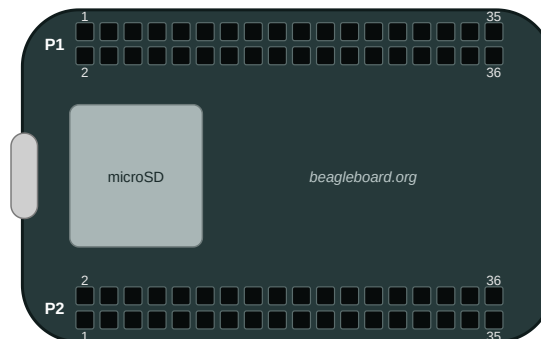
Step 2 of 4: power to each module

Red from each VCC or VIN column to the bottom + rail, black from each GND column to the bottom – rail. Read the module's letters before each wire.



board turned so P1 faces the breadboard: USB-C on the left, pin 1 at the left end of P1.

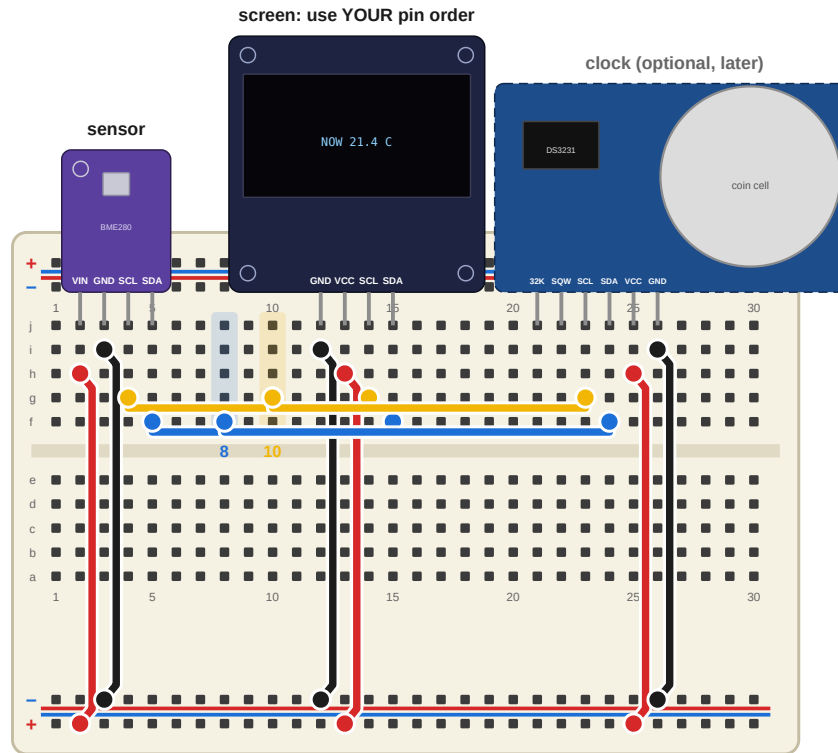
the rails: follow the printed red and blue lines on your own board; which one is outer varies.



Power uses the bottom rail pair, the one nearest the board. The wire leaves the module's column at row h or i and runs down past the groove to the rail. It only connects at its two ends.

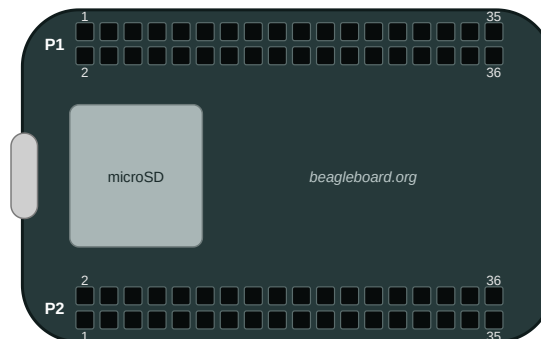
Step 3 of 4: the two shared columns

Column 8 collects every SDA (blue). Column 10 collects every SCL (yellow). Nothing plugs into 8 or 10 directly; they are meeting points.



board turned so P1 faces the breadboard: USB-C on the left, pin 1 at the left end of P1.

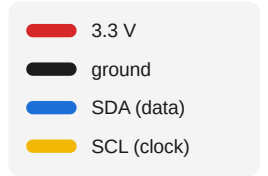
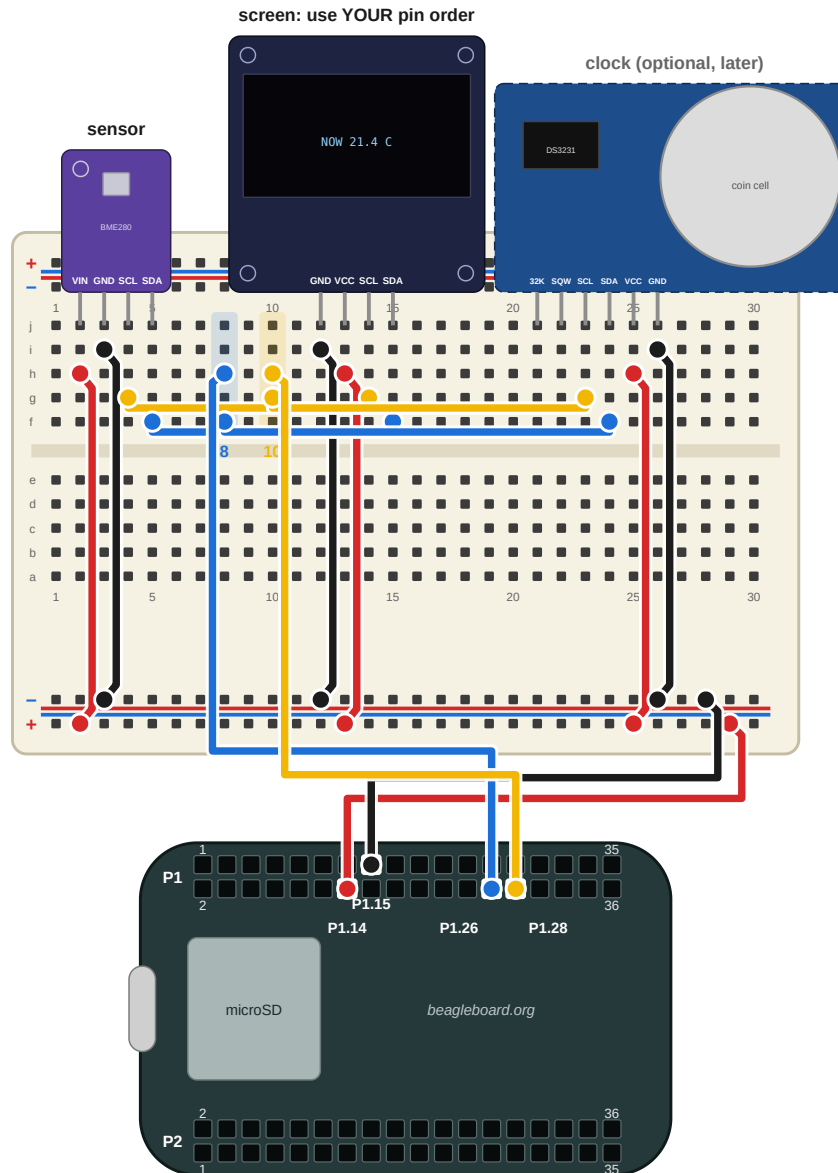
the rails: follow the printed red and blue lines on your own board; which one is outer varies.



Two blue wires end in column 8 and nowhere else. Two yellow wires end in column 10. A blue wire never touches a yellow column. Both columns sit between the sensor and the screen.

Step 4 of 4: four wires from the board

Red P1.14 to the + rail, black P1.15 to the – rail, blue P1.26 to column 8, yellow P1.28 to column 10. Then check every wire out loud.



board turned so P1 faces the breadboard: USB-C on the left, pin 1 at the left end of P1.

the rails: follow the printed red and blue lines on your own board; which one is outer varies.

Say it out loud: red leaves 14 and arrives at +. Black leaves 15 and arrives at –. Blue leaves 26 and arrives at column 8. Yellow leaves 28 and arrives at column 10.

The four wires, as a schematic

The same four wires for every module: power, ground, data, clock. This is the whole circuit.

